

STRESS AND STRAIN IN COMPOSITE LAMINATES

Part: **3.4 – STRENGTH OF COMPOSITE PARTS**

Chapter: **3401**

Title: **STRESS AND STRAIN IN COMPOSITE LAMINATES**

Revision: **1.3**

Date: **06/06/2024**

1 INTRODUCTION

1.1 OBJECT

Classical laminated plate theory reduces the stress/strain behaviour of a structural laminate to a solvable two-dimensional mechanics of deformable bodies problem. Fundamental relationship between in-plane forces and bending moments (per unit length) in the laminated-plate, and the deformation of its mid-plane surface (extensional strains and curvatures), is shown in figure below.

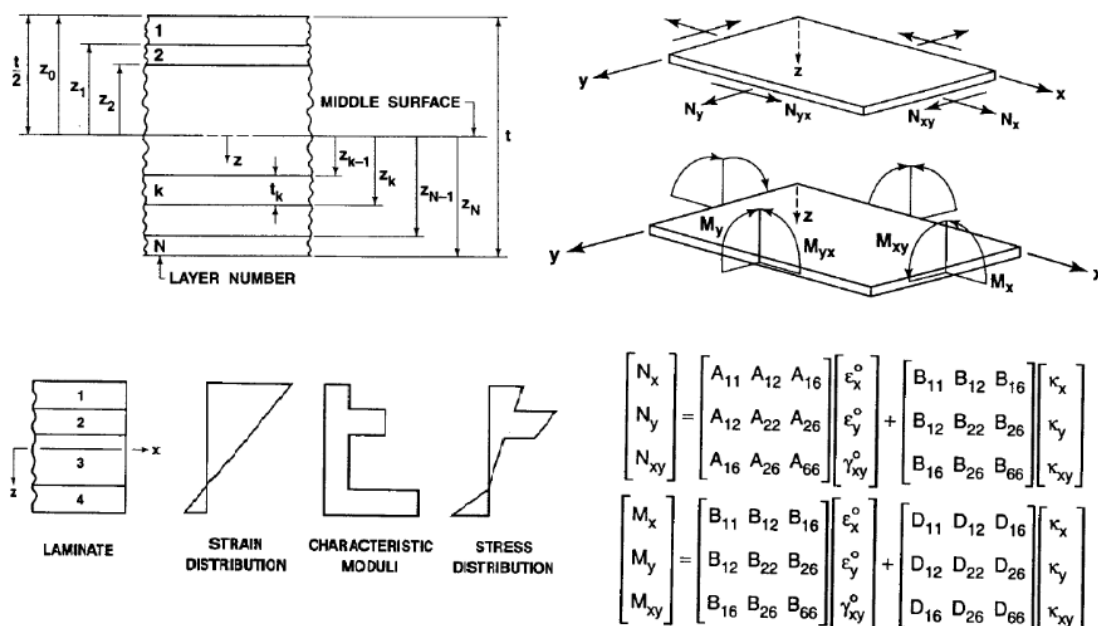


Figure 1. Basic equation of the Classical Laminate Theory, from Jones

The stiffness matrices $[A]$, $[B]$ and $[D]$ are obtained from the thickness (CPT) and elastic moduli of the ply (E_1 , E_2 , G_{12} , ν_{12}), and the ply angles of the stacking sequence. The details of the process can be found in typical bibliography. Theory shall not be treated here. Find suggested bibliography to review, among others, in the list of references of appendix.

1.2 SCOPE AND LIMITATIONS

This chapter is just a reminder of some basic relationships between stress and strain of composite laminates, according to the Composite Laminates Theory.

2 CHEATSHEET FOR BASIC ANALYSIS OF LAMINATES

Apparent/equivalent elastic moduli

The usage of the apparent/equivalent elastic moduli of the laminate is often advantageous. For a laminate with a common symmetric and balanced stacking sequence, these properties can be obtained from the in-plane stiffness matrix $[A]$ as:

$$\begin{aligned} E_x &= 1 / ([A^{-1}]_{11} \cdot t) = (A_{11} - A_{12}^2 / A_{22}) / t \\ E_y &= 1 / ([A^{-1}]_{22} \cdot t) = (A_{22} - A_{12}^2 / A_{11}) / t \\ \nu_{xy} &= -E_x \cdot t \cdot [A^{-1}]_{12} = A_{12} / A_{22} \\ \nu_{xy} / E_x &= \nu_{yx} / E_y \quad \rightarrow \quad \nu_{yx} = \nu_{xy} \cdot E_y / E_x = A_{12} / A_{11} \\ G_{xy} &= 1 / ([A^{-1}]_{66} \cdot t) = A_{33} / t \end{aligned}$$

In case of an unbalanced laminate the equivalent properties can obviously not capture the extension-shear coupling, but for certain applications the usage of equivalent properties can be meaningful. For unbalanced, yet symmetric laminates the properties can still be obtained using only the in-plane stiffness matrix $[A]$:

$$\begin{aligned} E_x &= \frac{1}{[A^{-1}]_{11} \cdot t} = \frac{A_{11}}{t} + \frac{A_{12}}{t} \left(\frac{A_{26}A_{16} - A_{12}A_{66}}{A_{22}A_{66} - A_{26}^2} \right) + \frac{A_{16}}{t} \left(\frac{-A_{16}}{A_{66}} + \frac{A_{26}A_{12}A_{66} - A_{26}^2A_{16}}{A_{22}A_{66}^2 - A_{26}^2A_{66}} \right) \\ E_y &= \frac{1}{[A^{-1}]_{22} \cdot t} = \frac{A_{22}}{t} + \frac{A_{12}}{t} \left(\frac{A_{26}A_{16} - A_{12}A_{66}}{A_{11}A_{66} - A_{16}^2} \right) + \frac{A_{26}}{t} \left(\frac{-A_{26}}{A_{66}} + \frac{A_{16}A_{12}A_{66} - A_{16}^2A_{26}}{A_{11}A_{66}^2 - A_{16}^2A_{66}} \right) \\ G_{xy} &= \frac{1}{[A^{-1}]_{66} \cdot t} = \frac{A_{66}}{t} - \frac{A_{26}^2}{A_{22} \cdot t} + \frac{2A_{12}A_{16}A_{26}A_{22} - A_{12}^2A_{26}^2 - A_{16}^2A_{22}^2}{t(A_{11}A_{22}^2 - A_{12}^2A_{22})} \\ \nu_{xy} &= -E_x \cdot t \cdot [A^{-1}]_{12} = \frac{\left(A_{12} - \frac{A_{16}A_{26}}{A_{66}} \right)}{\left(A_{22} - \frac{A_{26}^2}{A_{66}} \right)} \\ \nu_{yx} &= -E_y \cdot t \cdot [A^{-1}]_{12} = \frac{\left(\frac{A_{16}A_{26}}{A_{66}} - A_{12} \right)}{\left(\frac{A_{26}^2}{A_{66}} - A_{11} \right)} \end{aligned}$$

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For a laminate which is also unsymmetric, the stiffness matrix $[B]$, defining the bending-extensional coupling, shall not be null, and the laminate behaviour under load shall need the full equations system in Figure 1 to be explained. Even abovementioned expressions for elastic moduli shall still provide a valid estimation of the membrane stiffness, the “elastic axis” of the laminate will not be any more at the mid-plane surface, and bending-extensional coupling effects will occur when referencing loads to that reference system. See discussion on “Generalized bending stiffness” and “Laminated beams” later on in this same document.

Laminate elastic moduli and in-plane strain in quasi-orthotropic laminates

For standard designs, with symmetric-balanced laminates, the bending-extension coupling disappears (matrix $[B]$ is 0), as well as the shear-extension coupling ($A_{16}=A_{26}=0$). For these, so called quasi-orthotropic laminates, the equations above become simpler, and can be even expressed in terms of a more familiar formulation using the apparent elastic moduli of the laminate.

Extensional strains in quasi-orthotropic laminates:

$$\epsilon_{xx} = N_x / (E_x \cdot t) - N_y \cdot \nu_{xy} / (E_x \cdot t)$$

$$\epsilon_{yy} = N_y / (E_y \cdot t) - N_x \cdot \nu_{yx} / (E_y \cdot t)$$

$$\gamma_{xy} = N_{xy} / (G_{xy} \cdot t)$$

The relationship between the engineering shear strain γ_{xy} and the tensorial strain ϵ_{xy} (used alternatively in some methodologies and software) is simply:

$$\gamma_{xy} = 2 \cdot \epsilon_{xy}$$

Typical laminates include plies at 0° , 90° , 45° and -45° . Provided material 0° -direction is parallel to analysis reference X-direction it may be useful to have at hand expressions to get the extensional strains in $\pm 45^\circ$ directions:

$$\epsilon_{45} = (\epsilon_{xx} + \epsilon_{yy} + \gamma_{xy}) / 2$$

$$\epsilon_{-45} = (\epsilon_{xx} + \epsilon_{yy} - \gamma_{xy}) / 2$$

Generalized/equivalent bending stiffness

Literature offers a “corrected” equivalent/generalized bending stiffness, if the laminate is only slightly unsymmetric/unbalanced, in order to deal with problems like buckling analysis, avoiding the complexity of coupled bending/membrane equations:

$$[D^*] = [D] - [B][A]^{-1}[B]$$

CMH-17, Vol. 3, formula 9.2.2.1(b)

Laminated beams

Abovementioned equivalent bending stiffness matrix is independent of the reference plane the rest of the laminate stiffness matrices are referred to. If $[A]$, $[B]$, $[D]$ are the standard matrices referenced to the mid-plane surface, at an offset distance d , the corresponding matrices could be calculated as (see Ref. 3, chapter 6, subsection 6. “the parallel axis problem”, eq. 6.165):

$$[A'] = [A]$$

$$[B'] = [B] + d \cdot [A]$$

$$[D'] = [D] + d^2 \cdot [A] + 2 \cdot d \cdot [B]$$

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It is possible to check that the $[D^*]$ matrix will be exactly the same at this offset distance. The intention of recalling these expressions is, however, to show a method to find the distance d_x from the mid-plane surface to set the reference in a beam, of axis parallel to X, in which effects of axial forces and bending moments are uncoupled with un-symmetric laminates:

$$B'_{11} = 0 = B_{11} + d_x \cdot A_{11} \quad \rightarrow \quad d_x = -B_{11} / A_{11} \quad (\text{equivalent to eq. 6.174 in Ref. 3})$$

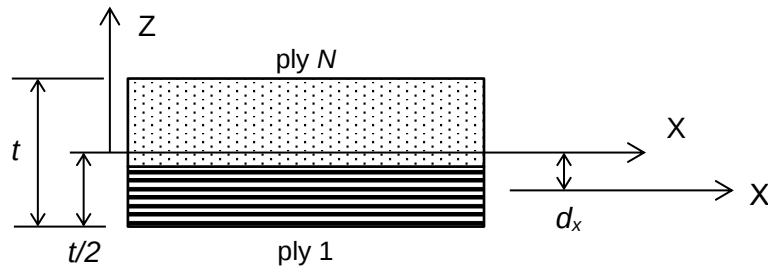


Figure 2. Translation of reference surface for uncoupling the effects of axial forces and bending moments in an unsymmetrically laminated beam of axis along X

That offset distance may be of interest for properly calculating the elastic axis/plane and inertias in beam sections composed by several segments of laminated composites.

Finally, formulas for deriving the bending moment of inertia of the laminated-beam (per unit width), and the apparent elastic flexural modulus are presented. For certain calculations, as inter-rivet buckling analysis, they can be useful. They can be obtained from the flexural stiffness matrix $[D]$ as:

$$EI_x = 1 / [D^{-1}]_{11}$$

$$EI_x = E_{f-x} \cdot t^3 / 12 \quad \rightarrow \quad E_{f-x} = 12 / (t^3 \cdot [D^{-1}]_{11}) \quad \text{CMH-17, Vol. 3, formula 8.3.3.2(c)}$$

If laminate is not symmetric, the equivalent bending stiffness matrix $[D^*]$ should be use instead.

In-plane stiffness matrix for $0^\circ, \pm 45^\circ, 90^\circ$ laminates

For symmetric-balanced laminates built on a plies stacking arranged according the typical base of angles $0^\circ, \pm 45^\circ$ and 90° it is possible to find more simple relationships

- ✓ From Jones' page 72, relationships between reduced stiffness of the lamina (0° material direction) and the engineering constants (elastic moduli) are:

$$\nu_{12} / E_1 = \nu_{21} / E_2$$

$$\nu_{12} = Q_{12} / Q_{22}$$

$$Q_{11} = E_1 / (1 - \nu_{12} \cdot \nu_{21})$$

$$(1 - \nu_{12} \nu_{21}) = Q_{12}^2 / (Q_{11} \cdot Q_{22})$$

$$Q_{22} = E_2 / (1 - \nu_{12} \cdot \nu_{21})$$

$$E_1 = Q_{12}^2 / Q_{22}$$

$$Q_{12} = \nu_{12} \cdot E_2 / (1 - \nu_{12} \cdot \nu_{21}) = \nu_{21} \cdot E_1 / (1 - \nu_{12} \cdot \nu_{21})$$

$$E_2 = Q_{12}^2 / Q_{11}$$

$$Q_{66} = G_{12}$$

$$G_{12} = Q_{66}$$

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Using condensed notation for the lamina stiffness matrix:

$$[Q_0] = [(Q_{11}, Q_{12}, 0), (Q_{12}, Q_{22}, 0), (0, 0, Q_{66})]$$

- ✓ Plies at 90° have the same stiffness matrix as 0° plies but interchanging Q_{11} and Q_{22} terms:

$$[Q_{90}] = [(Q_{22}, Q_{12}, 0), (Q_{12}, Q_{11}, 0), (0, 0, Q_{66})]$$

- ✓ From Jones' page 77, reduced stiffness for the plies-pairs at +45°/-45°...

$$Q'_{11} = Q'_{22} = 0.25 \cdot [Q_{11} + 2 \cdot (Q_{12} + 2 \cdot Q_{66}) + Q_{22}]$$

$$Q'_{12} = 0.25 \cdot (Q_{11} + Q_{22} - 4 \cdot Q_{66}) + 0.5 \cdot Q_{12}$$

$$Q'_{66} = 0.25 \cdot (Q_{11} + Q_{22} - 2 \cdot Q_{12} - 2 \cdot Q_{66}) + 0.5 \cdot Q_{66}$$

$$Q'_{16} = Q'_{26} = 0 \text{ (provided } +45^\circ \text{ and } -45^\circ \text{ plies come in pairs)}$$

$$[Q_{45}] = [(Q'_{11}, Q'_{12}, 0), (Q'_{12}, Q'_{22}, 0), (0, 0, Q'_{66})]$$

Then the reduced stiffness matrix for the laminate can be obtained weighting previous matrices by the respective weights w within the laminate of the plies at 0°, ±45° and 90° directions.

$$[Q] = [A] / t = w_0 \cdot [Q_0] + w_{45} \cdot [Q_{45}] + w_{90} \cdot [Q_{90}]$$

Obtaining the laminate $[A]$ stiffness matrix is immediate using the laminate thickness.

In-plane loads from a strain gauge rosette

Another typical application of the simplified theory for symmetric-balanced laminates is the obtention of the in-plane loads (N_x, N_y, N_{xy}) from the lectures of a rectangular strain gauge rosette, installed on a test specimen, with channels reporting strains at $X \parallel 0^\circ$, $Y \parallel 90^\circ$ and 45° directions.

The conversion of the lectures of a rectangular strain gauge rosette to the standard strain tensor is done using the following expressions:

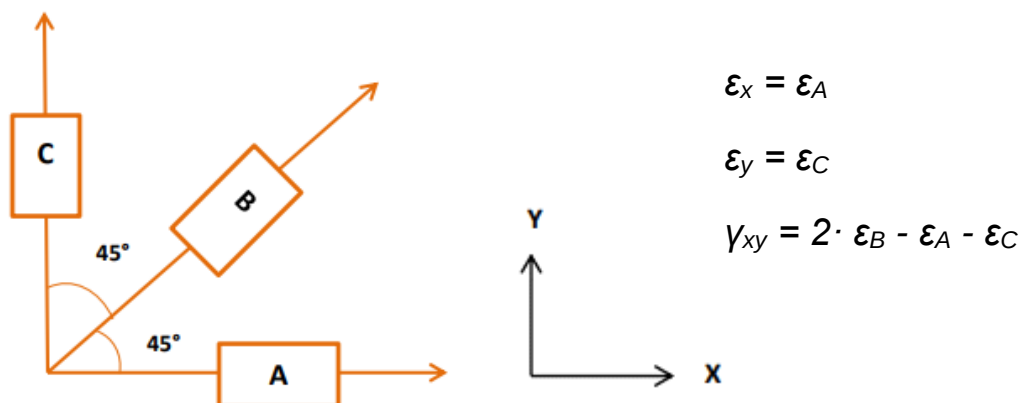


Figure 3. Transformation of strain gauge rosette strains to standard strain tensor

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Then, the calculation of the in-plane loads from the strain tensor, from the laminate apparent moduli and laminate thickness will be

$$[N] = [A] \cdot [\varepsilon] \quad \dots \text{ where: } [A] = t \cdot [Q] \quad \text{and} \dots$$

$$\begin{aligned} [\sigma] &= [Q] \cdot [\varepsilon] & \dots \text{with:} & [\sigma] = (\sigma_x, \sigma_y, \tau_{xy}) \\ [\sigma] \cdot t &= t \cdot [Q] \cdot [\varepsilon] & & [\varepsilon] = (\varepsilon_x, \varepsilon_y, \gamma_{xy}) \\ [N] &= [A] \cdot [\varepsilon] & & [N] = (N_x, N_y, N_{xy}) \\ & & & \nu_{xy} / E_x = \nu_{yx} / E_y \\ & & & Q_{11} = E_x / (1 - \nu_{xy} \cdot \nu_{yx}) \\ & & & Q_{22} = E_y / (1 - \nu_{xy} \cdot \nu_{yx}) \\ & & & Q_{12} = \nu_{xy} \cdot E_y / (1 - \nu_{12} \cdot \nu_{21}) = \nu_{yx} \cdot E_x / (1 - \nu_{xy} \cdot \nu_{yx}) \\ & & & Q_{66} = G_{xy} \end{aligned}$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
0	-	As subscript, relative to plies oriented at 0° in the laminate
45	-	As subscript, relative to plies oriented at ±45° in the laminate
90	-	As subscript, relative to plies oriented at 90° in the laminate
[A]	N/mm	In-plane stiffness matrix of the laminate
[B]	N	Membrane-bending coupling matrix of the laminate
CPT	mm	Cured Ply Thickness (thickness of the FRP ply)
[D]	Nmm	Bending stiffness matrix of the laminate
[D*]	Nmm	Equivalent/generalized bending stiffness matrix of the laminate
E ₁	MPa	Young's modulus of the FRP ply in warp direction
E ₂	MPa	Young's modulus of the FRP ply in weft direction
E _x	MPa	Apparent elastic modulus of the laminate in X direction
E _y	MPa	Apparent elastic modulus of the laminate in Y direction
EI	Nmm ²	Apparent bending stiffness of the laminated beam, per unit width
ε _x , ε _y	-	Extensional strains in the mid-plane surface of a laminate in X and Y directions (with super-index o in figure)
ε _{xy}	-	Shear tensorial strain
FRP	-	Fibre Reinforced Polymer
G ₁₂	MPa	Shear elastic modulus of the FRP ply in its plane
G _{xy}	MPa	Apparent shear elastic modulus of the laminate in its plane
γ _{xy}	-	Shear engineering strain in the mid-plane surface of a laminate in the X-Y plane (with super-index o in figure)
K _x , K _y , K _{xy}	mm ⁻¹	Curvature deformations in a laminate
M _x , M _y , M _{xy}	N	Moments per unit width
N _x , N _y , N _{xy}	N/mm	In-plane forces per unit width
[Q]	MPa	Stiffness matrix of the orthotropic lamina, or reduced stiffness matrix of the laminate
ν ₁₂	-	Poisson's ratio of the FRP ply
ν _{xy}	-	Poisson's ratio of the laminate
t	mm	Thickness of the laminate
w	%	Relative weight of plies in one direction, i.e. proportion of plies of the laminate oriented in one particular direction, i.e. at 0°, or at 90°, or at ±45° in this report, denoted by the subscript

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[Robert M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 2	[SAE] CMH-17	2012	Composite Materials Handbook (Volume 3)
Ref. 3	[S.W. Tsai, H.T. Hahn]	1980	Introduction to composite materials
Ref. 4	[Airbus DS] NT-T-ADS-95024	Is. A	Manual Teórico del Programa ROSETAS, versión 1.0

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	4.2.1	Detailed sizing of metallic and composite aircraft structures (module ORTHO-PROPERTIES)
ANCOLAB (Airbus DS)	4.2	Classical laminates analysis

Table 1. Software programs referenced in this chapter

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
1.0 01/08/2018	G. Baños	First issue	All pages
1.1 09/08/2018	G. Baños	Fix error in units of EI and terms of $[A]$ and $[Q]$ matrices	Page 4
1.2 20/07/2021	G. Baños F. Glock	Further explanations on apparent elastic modulus, matrix $[D^*]$ and laminated beams	Pages 2 - 6
1.3 06/06/2024	G. Baños	Added expressions for calculation of in-plane loads from a strain gauge rosette	Pages 5 - 6

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Methods	G. Baños 03/06/2024	-
Stress Analysis	-	F. Glock 06/06/2024

Table 3. Approval signatures table for last revision of this chapter